NPTEL ONLINE CERTIFICATION COURSES



Course Name: Software Project Management

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TOPIC: Project Estimation Techniques cont…



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CONCEPTS COVERED

- A taxonomy of estimating methods
- Size Estimation



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A taxonomy of estimating methods

- Top-down or Bottom-up - activity based, analytical
- Parametric or algorithmic models e.g. function points
- Expert Judgement - just guessing?
- Analogy - case-based, comparative
- Price to win





Pricing to win

- The project costs whatever the customer can spend on it.
- Advantages: You get the contract
- Disadvantages: Costs do not accurately reflect the work required. Either:
  - (1) the customer does not get the desired system or
  - (2) the customer overpays.





Pricing to win

- This approach may seem unethical and unbusiness-like….
  - However, when detailed information is lacking it may be the only appropriate strategy…
- Which is the most ethical approach?
  - The project cost is agreed on the basis of an outline proposal and the development is constrained by that cost
  - A detailed specification may be negotiated or an evolutionary approach used for system development





Project Parameters to be Estimated

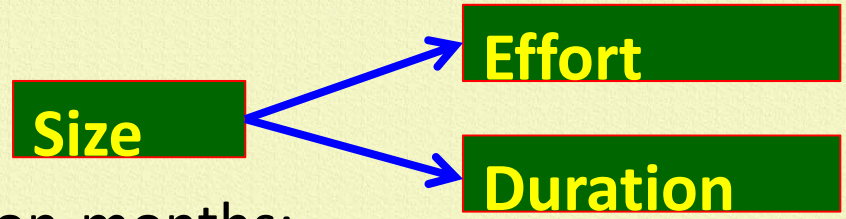
- For project planning, we need:
  - Effort (cost)
  - Duration
- Hard to estimate effort (or cost) or duration directly from a problem description.
- Effort and Duration can be measured in terms of project size (indirect metric)





Project Parameters to be Estimated cont …

- Size is a fundamental measure of work
- Based on the estimated size, two parameters are estimated:
  - Effort
  - Duration



- Effort is measured in person-months:
  - One person-month is the effort an individual can typically put in a month.





What is size? A Measure of Work…

- Project size is a measure of the problem complexity in terms of the effort and time required to develop the product.
- Two metrics are popularly used to measure project size:
  - Source Lines of Code (SLOC)
  - Function point (FP)
- SLOC is conceptually simple
  - But, FP is now-a-days favoured over SLOC
  - Because of the many shortcomings of SLOC.





Lines of code

- What's a line of code?
  - Originally proposed when programs were typed on cards with one line per card;
  - What happens when statements in Java span several lines or where there can be several statements on one line?
- What programs should be counted as part of the system? GUI? Built-in Class?
- Initially software development consisted of only writing code…





Lines of Code

- LOC ≡ Line of Code
- KLOC ≡
  - Thousands of LOC
- KSLOC ≡
  - Thousands of Source LOC
- NCKSLOC ≡
  - New or Changed KSLOC





LOC: Few things counter intuitive…

- The lower the level of the language, the more productive the programmer is:
  - The same functionality takes more code to implement in a lower-level language than in a high-level language.
- The more verbose the programmer, the higher the productivity:
  - Measures of productivity based on lines of code suggest that programmers who write verbose code are more productive than programmers who write compact code.





Major Shortcomings of SLOC

- Size can vary with coding style.
- Focuses on coding activity alone.
- Correlates poorly with quality and efficiency of code.
- Penalizes higher level programming languages, code reuse, etc.





Major Shortcomings of SLOC

- Difficult to estimate at start of a project from problem description
  - Only way to estimate is to make a guess…
  - So not useful for project planning
- Only a code measure
- Programmer-dependent
- Measures lexical/textual complexity only,
  - Does not consider structural or logical complexity.





Further Difficulties With SLOC

- SLOC can become ambiguous due to rapid changes in programming methodologies, languages, and tools:

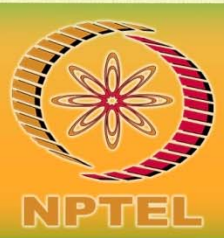
  -Language advancements

  -Automatic source code generation

  -Custom software and reuse

  - Object-orientation





Expert Judgment-based Techniques

1. Basic expert judgment
2. Weighted average estimating
3. Consensus estimating
4. Delphi





Basic Expert Judgment

- One or more experts predict software costs.
  - Process iterates until some consensus is reached.
- Advantages: Relatively simple estimation method. Can be accurate if experts have direct experience of similar systems
- Disadvantages: Very inaccurate if there are no experts available!





Basic Expert Judgment Method: Steps

- Coordinator presents each expert with a specification and an estimation form.
- Coordinator calls a group meeting in which the experts discuss estimation issues with the coordinator and each other.
- Experts fill out forms anonymously
- Coordinator prepares and distributes a summary of the estimation on an iteration form.
- Coordinator calls a group meeting, specially focusing on having the experts discuss points where their estimates varied widely.
- Experts fill out forms, again anonymously, and steps 4 and 6 are iterated for as many rounds as appropriate.

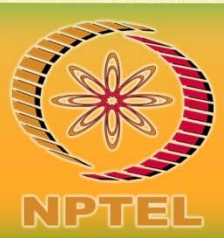




Basic Expert Judgement Method cont …

- An expert is familiar with and knowledgeable about the application area and the technologies
- Particularly appropriate where existing code is to be modified.
- Research shows that expert judgement in practice tends to be based on analogy…





Stages

- Identify significant features of the current project
- previous project(s) with similar features
- differences between the current and previous projects
- possible reasons for error (risk)
- measures to reduce uncertainty





Estimation by analogy

- The cost of a project is computed by comparing the project to a similar project in the same application domain:
- Advantages:
  - May be accurate if project data available and people/tools the same
- Disadvantages:
  - Impossible if no comparable project has been tackled.
  - Needs systematically maintained cost database

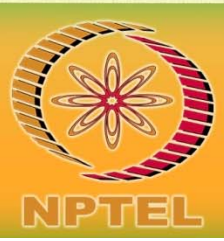




Basic Expert Judgement: Cons

- Hard to quantify
- It is hard to document the factors used by the experts or expert-group.
- Expert may be biased, optimistic, or pessimistic, even though they have been decreased by the group consensus.
- The expert judgment method always complements the other cost estimating methods such as algorithmic method.





Weighted average estimates

- Weighted average estimating is also known as sensitivity analysis estimating.
- Three estimates are obtained rather than one.
  - Best case (O = optimistic), worst case (P = pessimistic) and most likely (M = median).
  - This provides a more accurate estimate than when only one estimate is used.
- These are then used in the following formula:

Estimated effort = (O + 4M + P) / 6



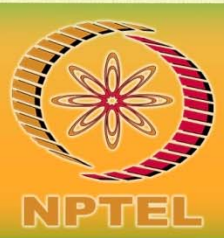


Consensus estimating

Steps in conducting a consensus estimating session:

- A briefing is provided to the estimating team on the project.
- Each person is provided with a list of work components to estimate.
- Each person independently estimates O, M and P for each work component.
- The estimates are written up on the whiteboard.
- Each person discusses the basis and assumptions for their estimates.
- A revised set of estimates is produced.
- Averages for the O, M and P values are calculated.
- These values are used in the formula.





Delphi Estimation

- A variation of consensus estimation technique
- Team of Experts and a coordinator.
- Experts carry out estimation independently:
  - mention the rationale behind their estimation.
  - coordinator notes down any extraordinary rationale:
    - circulates the estimation rationale among experts.
- Experts re-estimate.
- Experts never meet each other
  - to discuss their viewpoints.





Delphi

- Delphi is an expert survey in two or more "rounds".
- Starting from the second round, a feedback is given (about the results of previous rounds).
- The same experts assess the same matters once more - influenced by the opinions of the other experts
- **important: anonymity**

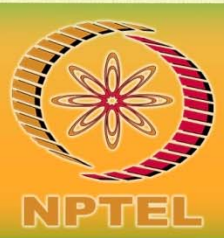




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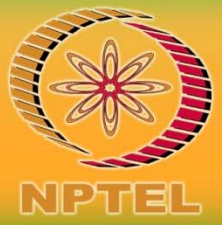




Summary

- Discussed the Price to win estimation method.
- Presented Size estimation.
- Also, discussed Expert Judgement – based estimation techniques such as
  - Basic judgement
  - Weighted average estimating
  - Consensus estimating
  - Delphi
- Explained the Analogy based estimation.





References :

1. B. Hughes, M. Cotterell, R. Mall, *Software Project Management*, Sixth Edition, McGraw Hill Education (India) Pvt. Ltd., 2018.
2. R. Mall, *Fundamentals of Software Engineering*, Fifth Edition, PHI Learning Pvt. Ltd., 2018.






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